

Major Advanced Topics

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Multiphoton Resonance Meets Tunneling Ionizing: High-Efficient Photoexcitation in Strong-Field-Dressed Ions

(Phys. Rev. Lett. 133, 113201 (2024))

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Summary

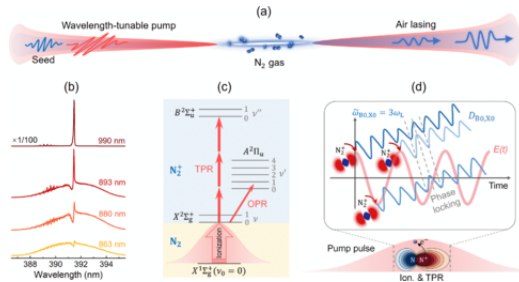
- Study investigates how tunneling ionization helps leverage multiphoton excitation to achieve inversion of populations in ionic gases
- The target is N_2^+ obtained from molecular N_2 and ionized with a strong field
- The researchers showed that by exploiting techniques as phase-locking and stimulated emission, we can maximize ionic the lasing of the target by one order of magnitude
- This is another study deepening our understanding of strong field optics

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Physical mechanism of tunnel ionization, along with photoexcitation with TPR

- Fig. 1(a) : Schematic of the experiment setup
- Fig. 1(b) : Typical emission spectra for different wavelengths
- Fig. 1(c) : Energy-level diagram to ionization and photoexcitation
- Fig. 1(d) : Physical mechanism of high-effiecient photoexcitation with coexisting OPR and TPR
 - Photons are emitted at each half period of the field
 - Notice their constructive interference in Fig. 1(d)



The Fano amplitude are given by $\sigma(E) = |T(E)|^2$

$$T(E) = \langle X|V|B \rangle + \sum_{\eta} \frac{\langle X|V|\psi_{\eta} \rangle \langle \psi_{\eta}|V|B \rangle}{E - E_{\eta} + i\Gamma/2} \quad (1)$$

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The experimental setup, details

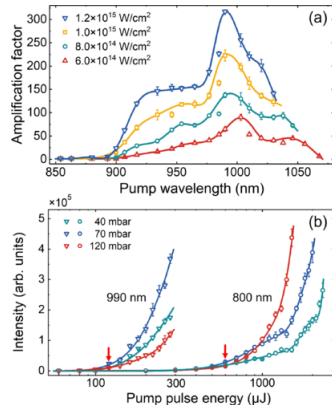
- The experimental uses two independent lasers
 - A wavelength-tunable pump ranging in 800 – 1070 nm, 60 fs
 - A seed (391 nm), there to stimulate/amplify the transition $B \rightarrow X$ between the two ionic states
- The initial ionization of N_2 is done through tunnel ionization, made possible because the field is strong ($I_p = 0.573$ a.u.)
- Subsequently,
 - 3 photons are absorbed (TPR) to reach $B^2\Sigma_u^+$ in some vibrational state ν'' , or
 - 1 photon is absorbed (OPR) to reach $A^2\Pi_u$ in some other vibrational state ν'
- Use is made of phase-locking in order to optimize photoexcitation and boots efficiency

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Lasing intensity

- Fig. 2(a) : Amplification factor as a function of the pump intensity
 - Amp. factor is the ratio of the spectral intensity of N_2 lasing to that of the seed (391 nm)
 - Gradual increase with the pump intensity
 - Maximum stuck to (990 nm) for sufficient intensities
- Fig. 2(b) : Dependence of the lasing intensity with the pump energy
 - Comparison of the 990 nm pump with 800 nm pump used in previous study
 - Different gas pressures are considered
 - 990 nm more efficient, requiring a less gas pressure and pump energy



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Theoretical insight

- From Lindbald theory, used to describe the dynamics of open systems, we write

$$\frac{d\rho}{dt} = -i[H, \rho] + \left[\sum \gamma \left(A\rho A^\dagger - \frac{1}{2}A^\dagger A\rho - \frac{1}{2}\rho A^\dagger A \right) \right], \quad (2)$$

hastily reduced to, the following form

$$\frac{d\rho}{dt} = -i[H, \rho] + L(\rho) \quad (3)$$

- The Lindbald superopareator $L(\rho)$ meant to model the environmental conditions that make up the open quantum system
- Here, it represents the population exchange from ionization and decay not captured by the hermitian Hamiltonian, thus

$$\frac{d\rho^+}{dt} = -i[H(t), \rho^+] + \left(\frac{d\rho_{n\nu}^+}{dt} \right)_{\text{ionize}} + \left(\frac{d\rho_{n\nu, n'\nu'}^+}{dt} \right)_{\text{decay}} \quad (4)$$

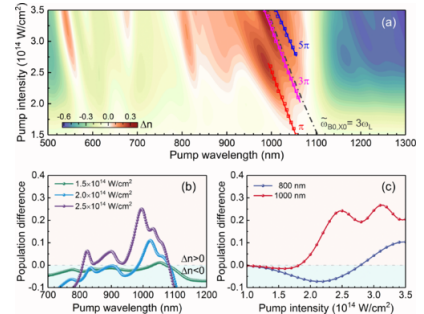
- Here we will be interested in $\Delta n = \rho_{B0, B0}^+ - \rho_{X0, X0}^+$

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Population differences

- Fig. 3(a) : Calculated Δn between $B(\nu'' = 0)$ and $X(\nu = 0)$ with pump wavelength and intensity
 - Red, magenta and blue lines indicate the 3 photons Rabi oscillations ending at $\phi = \pi, 3\pi, 5\pi$
 - Black dash-dot indicates the gain channel of TPR $\tilde{\omega}_{B0,X0} = 3\omega_L$
- Fig. 3(b) : Relation between Δn and the pump wavelength
 - Pop. inversion appearing in a larger and larger range with intensity
 - The peak stays around 1000 nm regardless
- Fig. 3(c) : Dependence of Δn on the intensity for 800 nm and 1000 nm pumping
 - Intensity required for pop. inversion much lower for 1000 nm

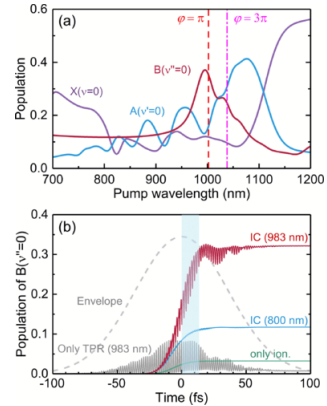


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Populations as a function of wavelength and time

- Fig. 4(a) : Populations in X , A , and B as a function of the pump wavelength
 - $I = 2.5 \times 10^{14} \text{ W/cm}^2$
 - $\rho_{X0,X0}^+$ decrease steadily due to OPR exc.
 - TPR excitation is very efficient around the TPR wavelength, even suprasing OPR
- Fig. 4(b) : Evolution of $\rho_{B0,B0}^+$ in different cases
 - Dashed line is the field envelop
 - When only considering ionization or TPR, excitation rates are 3.2% and 0.7%
 - When only considering both, excitation rate is 32.2%
- Thus, an efficient combination of tunnel ioniz. and TPR significantly improve photoexcitation rates



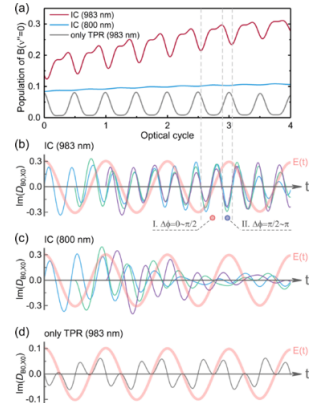
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Population dynamics

- Fig. 5(a) : Pop. dynamics of the $B(\nu'' = 0)$
- Excitation rates are proportional to $\text{Im}(D_{B0,X0}) E(t)$
- Fig. 5(b-d) : Evolution of the ionic dipoles periodically created for
 - (b) 983 nm, where the ionic dipoles are in phase
 - (c) 800 nm, they exhibit random phase
 - (d) only TPR

$$D_{B0,X0} = \langle B, \nu'' = 0 | \hat{d} | X, \nu = 0 \rangle \quad (5)$$



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Conclusion

This paper essentially shows that within a strong field,

- multiphoton excitation synergistically happens along tunnel ionization around the optimal wavelength pertaining to the system,
- their simultaneous involvement makes the ionic excited-state population be nearly 1 order of magnitude higher compared to their individual contribution
- there is a crucial role played by the ionic coherence and phase matching, revealing the interplay of tunneling and multiphoton resonance by the subcycle phase analysis